

Experiment :-

Study of LAN Topology. Network devices and Address Configuration.

Objective :-

- 1) To Study the LAN topology used in College Campus
- 2) To identify networking device & understand their functions.
- 3) To analyze wired & wireless Communication method used in the network.
- 4) To measure Speed of network for wired & wireless connections.
- 5) To obtain MAC, IP Address & Subnet mask

Problem Statement - 1

Draw & identify LAN topology of College Campus & Clearly label all components such as router, switches & etc.

Solⁿ :-

The College Campus network observed this study consists of:

- * Layer-3 Core Switches in the Server room.
- * Managed by PoE+ Switches.
- * Wireless Access point (APs)
- * Ethernet port in Classroom & Laboratories.

Network Topology :-

- (i) Core layer
- (ii) Distribution layer
- (iii) Access layer

These all are the network hierarchical layer.

Hybrid Hierarchical Topology which combines.

- (i) Tree topology
- (ii) Extended Star topology
- (iii) Ring Topology (Server room).

Core layer :-

- Core layer contains Server Room and Server room have Firewall, Router, 2-23+ Switches patch Rack, Server, inverter etc.

Distribution layer :-

- Distribution layer contains Server Room and
- The 4 Switches are-

- (i) Main Build $\rightarrow R_7$ (Room no-213, 2nd Floor)
- (ii) Main Build $\rightarrow R_1$ (Room no-301, 3rd Floor)
- (iii) Main Build $\rightarrow R_6$ (Room no-113, 1st Floor)
- (iv) Main Build $\rightarrow R_2$ (Room no-201, 2nd Floor)

- This 4-Switches contains $\rightarrow$

- (i) 18 port Gigabit Easy Smart Switch with 16-port TL-SG1218-P1

- (ii) SG 24 28xMP $\rightarrow$ 24 port Gigabit & 4-port 10G SFP+, L2+ Managed Switches with 24 port POE+

$\rightarrow$ Now let's discuss these 2 above components $\rightarrow$

Now TP links TL-SR1218MPE

- likely used for: CCTV Camera, wifi access point etc.

- 18 port Gigabit "Easy Smart" Switch
- 16 port Support POE+

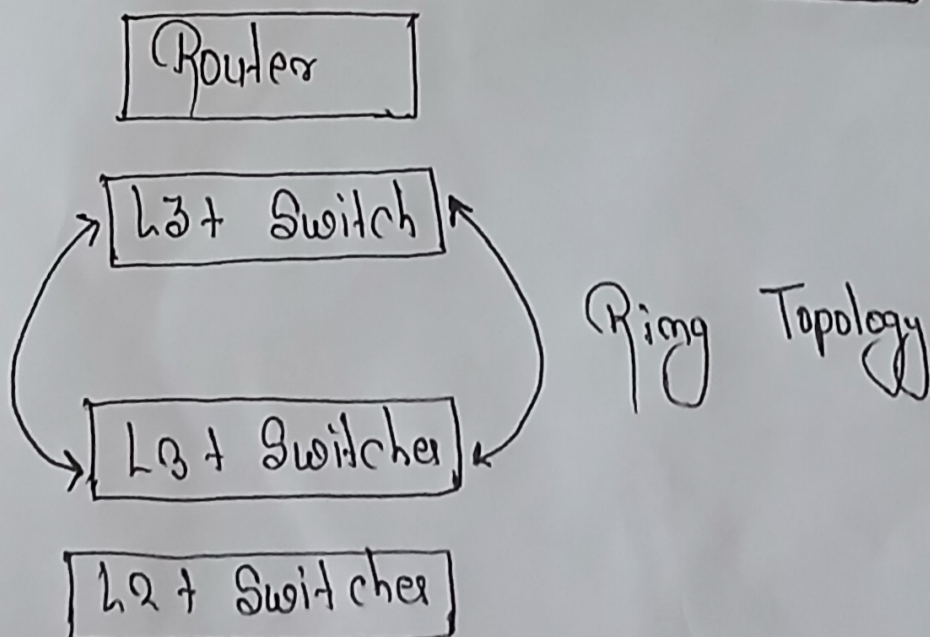
(2) TR - Link Canada 5934-28xMP

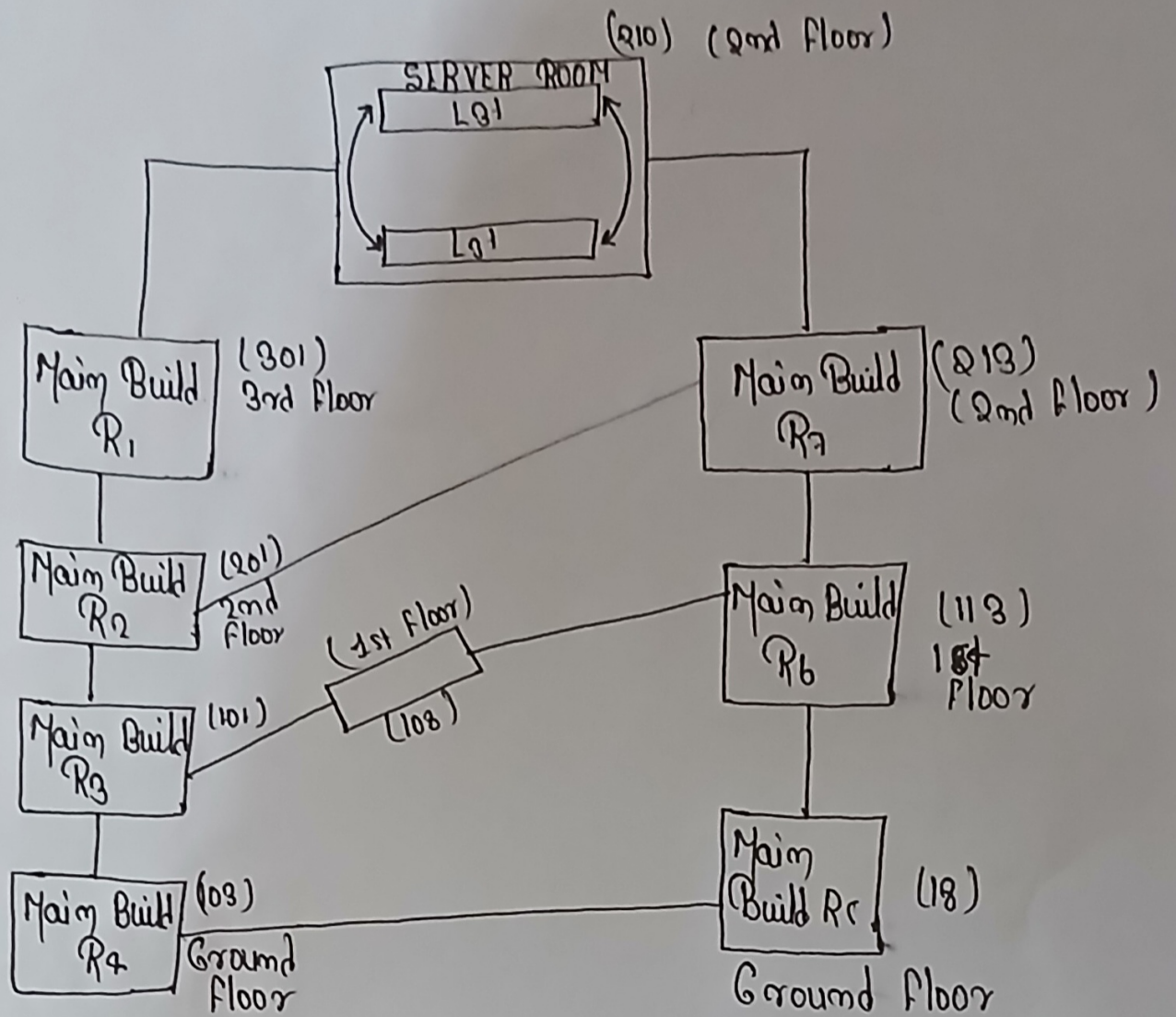
- Higher-end managed Layer 2+ Switch
- 24 POE + Port, uplink, 10G SFP
- usually acts as a main distribution for each floor

Access Layer :-

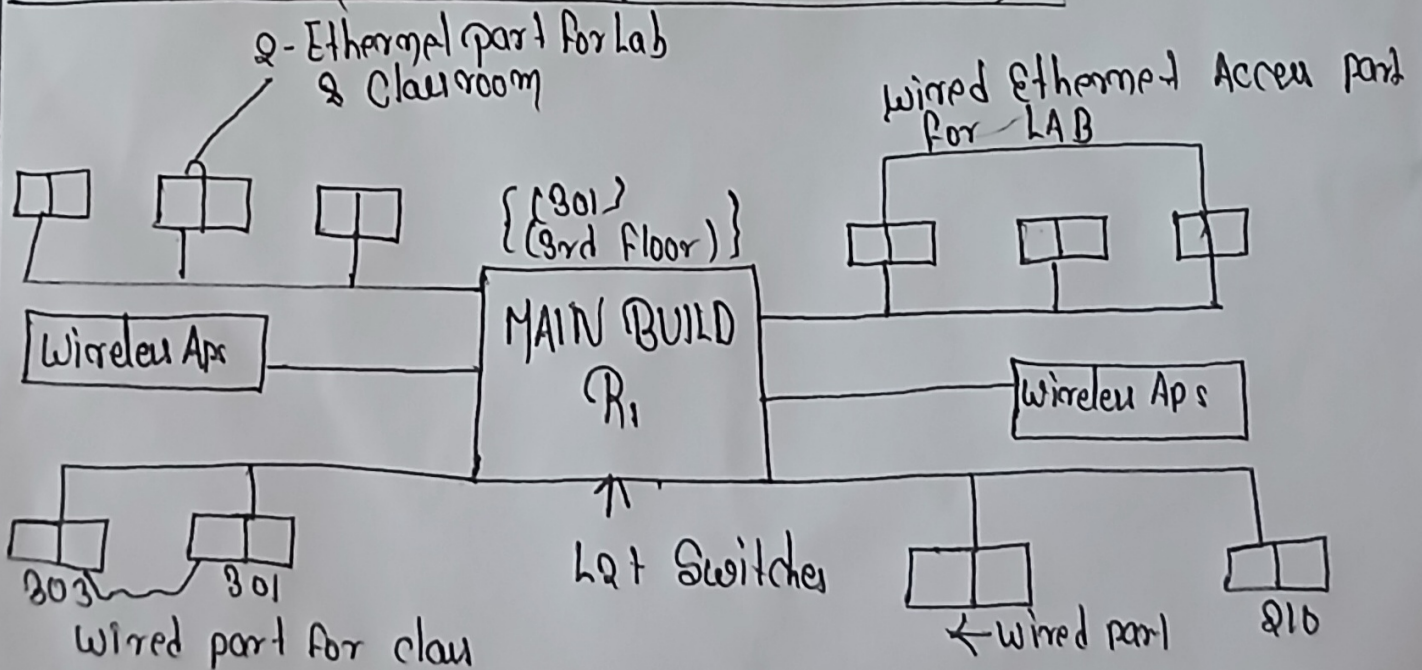
- Access layer contains 3-switches i.e.
 - (i) Main Build R₁ - [Room no-301, 3rd Floor]
 - (ii) Main Build R₄ - [Room no - Ground Floor]
 - (iii) Main Build R₅ - [Room no-18, Ground Floor]
- Every Switches act as access layer which helps to provide network connectivity to all Classroom, Lab, Faculty Room etc.
- Access Layer directly Connect :-
 - (i) Ethernet Port
 - (ii) Wireless AP
 - (iii) Lab System
 - (iv) Outdoor Wireless AP (ex-D18)

→ Server Room Architecture / Topology :-





- R_1, R_7, R_2, R_6 → Distribution layer
 - Server Room → Core layer
 - $R_1, R_2, R_3, R_4, R_5, R_6, R_7$ → Access layer
- Access Layer SWITCH (3rd Floor - 301)



```
Windows PowerShell
PS C:\Users\SAKSHI KUMARI> netsh wlan show interfaces

There is 1 interface on the system:

Name                : Wi-Fi
Description          : Realtek 8852BE-VT Wireless LAN WiFi 6 PCI-E NIC
GUID                : 763667f0-2ac0-47e8-96fb-c6665f7deea6
Physical address     : 38:8d:3d:b9:22:ec
Interface type       : Primary
State                : connected
SSID                : GECK-Campus-WiFi
AP BSSID             : a8:6e:84:cb:36:c8
Band                 : 2.4 GHz
Channel              : 1
Connected Akm-cipher : [ akm = 00-0f-ac:02, cipher = 00-0f-ac:04 ]
Network type         : Infrastructure
Radio type           : 802.11n
Authentication       : WPA2-Personal
Cipher               : CCMP
Connection mode      : Auto Connect
Receive rate (Mbps)  : 144.4
Transmit rate (Mbps) : 144.4
Signal               : 80%
Rssi                 : -67
Profile              : GECK-Campus-WiFi
QoS MSCS Configured  : 0
QoS Map Configured   : 0
QoS Map Allowed by Policy : 0

PS C:\Users\SAKSHI KUMARI> |
```

```
PS C:\Users\SAKSHI KUMARI> ipconfig
```

Windows IP Configuration

Unknown adapter McAfee VPN:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
```

Wireless LAN adapter Wi-Fi 2:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
```

Wireless LAN adapter Wi-Fi 4:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
```

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::98de:9f9f:f1f8:abe6%14
IPv4 Address. . . . . : 10.85.21.245
Subnet Mask . . . . . : 255.255.248.0
Default Gateway . . . . . : 10.85.16.1
```

Ethernet adapter Bluetooth Network Connection:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
```

```
PS C:\Users\SAKSHI KUMARI> |
```

```
PS C:\Users\SAKSHI KUMARI> getmac
```

Physical Address	Transport Name
00-FF-81-FD-2E-77	Media disconnected
38-8D-3D-B9-22-ED	Media disconnected
38-8D-3D-B9-22-EC	\Device\Tcpip_{763667F0-2AC0-47E8-96FB-C6665F7DEEA6}

→ Difference b/w wired & wireless Speed?

Wired Network	Wireless Network
• Faster Communication • Stable Connection • Low latency • Dedicated physical medium	• Slower Communication • Affected by distance & Interference • Higher latency • Shared radio medium

→ Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance.

→ Wireless connections use radio waves which are affected by interference, distance, obstacles & multiple connected users.

Q. Role of Subnet mask in network?

A Subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

• It is important for:

- (i) Network Segmentation
- (ii) Efficient routing
- (iii) Communication b/w devices.

OBSERVATION

During the study of the college network infrastructure:-

- (i) A Centralized Server room was used as network backbone.
- (ii) Two L3 Switches were connected in redundant cyclic structure.

(iii) Managed TP-link POE + switches were used in network cabinets.

Advantage of Existing Network?

- Scalable architecture
- Centralized Management
- Supports POE + device
- Easy expansion of floors & classroom.
- Support both wireless & wired.
- Redundant link for failover.

Conclusion

The observed Campus LAN follow a Hybrid hierarchical topology consist of tree, Extended Star, Ring. The network use Manage of POE Switches wireless access point - Ethernet Port & Centralized Server/Storage Structure to provide efficient wired & wireless communication, across the Campus.